Upper Limb Neurological Exam — OSCE Checklist

Deakin MD2 · BB6/BB7 · Time: 8–10 min

Tick each step as you perform or verbalise it. **Shaded rows (*)** *are optional or done if indicated. Order:* **Inspection → Tone → Power → Reflexes → Coordination → Sensation**. *Always compare* **left vs right** *and narrate aloud.*

1. Introduction (WIPER)

☐	**W**ash hands / hand hygiene
☐	**I**ntroduce self (name, role)
☐	**P**atient ID: name and DOB
☐	**E**xplain the exam and gain consent
☐	**E**xpose: top half exposed (singlet / sports bra); maintain dignity, good lighting
☐	Ask about **pain** before touching
☐	Equipment ready: tendon hammer, cotton wool, neurotip (sharps safety, gentle on thin skin), 128 Hz tuning fork

2. Examination

	Step	What to do / say	Root / abnormal
Inspection			
☐	General	Posture, walking aids, scars, skin changes	Hemiplegic posture, deformity
☐	Muscle bulk	Both sides: shoulder girdle, arms, thenar/hypothenar eminences, dorsal interossei	Wasting = LMN; symmetry
☐	Fasciculations	Watch muscles at rest for 10–20 s	LMN (e.g. MND)
☐	Abnormal movements	Tremor (rest/action), myoclonus, chorea, dystonia	Resting tremor = Parkinson's
☐	Pronator drift	"Arms out, palms up, eyes closed" for 20–30 s	Pronates + drifts down = UMN
Tone			
☐	Wrist + elbow	Ask about pain; patient relaxed; passively flex/extend and pronate/supinate at variable speed	Normal / increased / decreased
☐	Rigidity	Feel for cogwheeling (+ tremor) or lead-pipe rigidity	Extrapyramidal (Parkinson's)
☐	Spasticity	Velocity-dependent catch (clasp-knife)	UMN
Power (MRC 0–5) — isolate the joint, match the patient's strength			
☐	Shoulder abduction	"Lift your arms up like wings, don't let me push down"	C5, C6 — deltoid, supraspinatus
☐	Shoulder adduction	"Push your arms down to your sides"	C6, C7, C8 — pec major, lat dorsi
☐	Elbow flexion	"Pull your arm in towards you"	C5, C6 — biceps, brachialis
☐	Elbow extension	"Push me away"	C7, C8 — triceps
☐	Wrist flexion	"Push your wrists down, don't let me push them up"	C6, C7 — FCR, FCU
☐	Wrist extension	"Push your wrists up, don't let me push them down"	C7, C8 — ECR, ECU
☐	Finger extension	"Straighten your fingers, don't let me bend them"	C7, C8 — extensor digitorum
☐	Finger flexion	"Squeeze my two fingers"	C7, C8 — FDP, FDS
☐	Finger abduction	"Spread your fingers, don't let me push them together"	C8, T1 — dorsal interossei
☐	Finger adduction	Resist examiner separating fingers (paper grip)	C8, T1 — palmar interossei
☐	Thumb opposition*	"Touch your thumb to your little finger"	Median nerve (T1)
Reflexes — relaxed, positioned; reinforce (clench teeth) if absent			
☐	Biceps	Arm in lap, forearm pronated; finger on tendon, strike finger	C5, C6
☐	Brachioradialis	Fingers on lateral radius just above wrist; strike	C5, C6
☐	Triceps	Arm loosely supported across chest; strike tendon above olecranon	C7, C8
☐	Finger jerk*	Hand supinated, fingers slightly flexed; strike your own fingers on theirs	C8
☐	Hoffmann's*	Flick distal phalanx of middle finger; watch thumb	Thumb flexes = UMN
Coordination			
☐	Finger–nose	"Touch my finger, then your nose", move target; repeat each side	Past-pointing, intention tremor
☐	Rapid alternating	Pat palm with front then back of the other hand, fast	Dysdiadochokinesia
☐	Rebound	Outstretched arms, eyes closed; tap down — arm returns	Overshoot = cerebellar
Sensation — demonstrate on sternum first, eyes closed, compare sides			
☐	Light touch	Cotton wool: dab each dermatome C5–T1	Dermatomal / glove pattern
☐	Pain	Neurotip: sharp or blunt? Same dermatomes	Spinothalamic loss
☐	Temperature*	Cold tuning fork if indicated	Spinothalamic
☐	Vibration	128 Hz fork on DIP; if absent move to MCP → wrist → elbow	Dorsal column loss
☐	Proprioception	Hold DIP by its sides; "Up or down?"; if absent move proximally	Dorsal column / sensory ataxia

3. Closing and presenting

☐	Thank patient, help them dress, wash hands
☐	Completion statement: "To complete, I would examine the lower limbs and cranial nerves, and check the neck/spine."
☐	Present one finding per domain: "On inspection normal bulk with no fasciculations; tone normal; power 5/5 throughout; reflexes normal and symmetrical; coordination normal; sensation intact. Overall a normal upper limb neurological exam."

4. Quick reference

MRC	Power grade
0	No contraction
1	Flicker, no movement
2	Moves with gravity eliminated
3	Moves against gravity, not resistance
4	Against resistance, weaker than normal
5	Normal power

Dermatome	Test site
C5	Shoulder tip / lateral upper arm
C6	Lateral forearm, thumb
C7	Middle finger
C8	Little finger
T1	Medial upper arm / elbow
Reflexes	0 absent · 1+ reduced · 2+ normal · 3+ brisk · 4+ clonus

Sign	UMN	LMN
Bulk	Normal / disuse	Wasting
Fasciculations	No	Yes
Tone	Increased (spastic)	Decreased
Power	Pyramidal weakness (extensors)	Weak in root/nerve distribution
Reflexes	Brisk, Hoffmann's +	Reduced / absent

Nerve	Classic finding
Radial	Wrist drop; sensory loss first dorsal web space
Ulnar	Claw hand (4th/5th), interossei wasting, Froment's +
Median	Thenar wasting, weak opposition; Tinel's / Phalen's (carpal tunnel)

- **Pyramidal pattern (UMN):** extensors weaker than flexors in the arm — shoulder abduction, elbow and wrist extension go first.
- **Cerebellar signs (DANISH):** Dysdiadochokinesia, Ataxia, Nystagmus, Intention tremor, Slurred speech, Hypotonia — ipsilateral to lesion.
- **Glove distribution** sensory loss = peripheral neuropathy (diabetes, B12, alcohol); **dermatomal** = root; **nerve territory** = mononeuropathy.
- **Pitfalls:** overpowering the patient, not isolating the joint, triceps tested with arm unsupported, skipping reinforcement, testing sensation with eyes open.